

Final Project(PULPNET): Build a Transformer-Based Chatbot

In this project, you are required to design and deploy an intelligent chatbot using an open-source transformer model. The primary aim is to introduce you to the development pipeline of modern AI-driven conversational systems while applying your skills in data scraping, model integration, and deployment.

Objective

The chatbot should be capable of answering user queries based on information extracted from official or student-led IIT Kanpur web sources. It will serve as a practical application of language models and can assist in delivering relevant information to users in a conversational format.

Data Collection

You are free to scrape and use content from any IITK-related website. Recommended sources include the Vox Populi website, ICS website, and faculty profile pages or departmental web portals. Ensure that the data is cleaned appropriately, removing HTML tags, special characters, and unnecessary formatting. Organize the text in a structure suitable for question-answering or summarization tasks.

Model Selection and Integration

You could use an open-source transformer model from libraries like Hugging Face's `transformers`. Preferably use pre-trained models such as BERT, RoBERTa, DistilBERT, LLaMA, or TinyLlama, depending on the computational resources available. You may use the model in a zero-shot/inference-only manner by embedding the context and querying it dynamically. We are open to innovation.

Chatbot Development

Develop a chatbot interface capable of processing user input and generating relevant responses based on your selected model and dataset. It is fine if it takes only one question and generates the answer for it.

Deployment with Streamlit

Deploy the chatbot using Streamlit. The interface should include a text input box for users to type their queries and an area to display model responses.

Video Demonstration

Along with the deployment, you are also required to record a short video (under 1 minute) demonstrating your chatbot in action. The video should clearly show the interface, a few example queries, and the corresponding responses generated by the chatbot. This will help in evaluating the usability and completeness of your implementation.

Deliverables and Evaluation

Your submission must be pushed into your GitHub repository that you submitted and must include the following:

- Your source code, processed data, and a README explaining how to run the project.
- A public or shareable link to your Streamlit app (or instructions to run it locally) in the README file.
- A video file (max 1 minute) demonstrating the chatbot functionality.

Evaluation will be based on the creativity in data sourcing, effectiveness of the chatbot's responses, user interface quality, completeness of deployment, and adherence to submission guidelines.

Deadline

The deadline to submit the project is 5th July 2025 .